

Structural Diagnosability of Enterprises as Continua: Formal Definition and Decidability

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Abstract

We address a single frozen scientific question: Is *structural diagnosability* a formally well-defined and decidable property of enterprises modeled as continua (K_{EA}), independent of managerial intent and KPI observability?

This preprint provides a formal definition of structural diagnosability strictly within the immutable OC core and the executable-closed enterprise continuum specification (ETS.K_EA.v1.1). We then reduce diagnosability to a finite decision problem over the enterprise continuum representation, yielding a decidability result under explicitly stated minimal assumptions already present in the specification.

Keywords. structural diagnosability; enterprise continuum; formal methods; decidability; OC; K_{EA}

1 Problem and Scope

1.1 Frozen scientific question

This preprint answers exactly one question (Q1):

“Is structural diagnosability a formally well-defined and decidable property of enterprises modeled as continua (K_{EA}), independent of managerial intent and KPI observability?”

1.2 Hard scope constraints

We enforce the following constraints:

- No new primitives beyond OC core and ETS.K_EA.v1.1.
- No reinterpretation of OC or ETS.K_EA.v1.1.
- No methodological or prescriptive content (no roadmaps, no consulting, no advice).
- No weakening of STOP / No-Go semantics and invariants.
- The paper stands alone; it does not require reading any whitepaper.

1.3 Operational meaning

The term ‘structural’ in this paper refers to structure encoded in the enterprise continuum formalism (state space, boundaries, axes, thresholds, flows, cycles, and continuity). The term ‘diagnosability’ refers to whether a decision procedure can determine, from a formally admissible description of an enterprise-as-continuum, whether the continuum is diagnosable in the sense defined in section 3.

We explicitly do *not* assume the availability of KPIs, managerial intent, or any specific human interpretation layer. Any observation model, if used, is introduced only as a derived construct definable from the immutable primitives.

2 Formal Setting

We work strictly within the OC core primitives and the enterprise-continuum specification K_{EA} as fixed by `ETS.K_EA.v1.1`. The enterprise continuum is treated as a formal object equipped with (at minimum) the following immutable components:

$$(\Omega(K_{EA}), \partial\Omega(K_{EA}), A(K_{EA}), P(K_{EA}), \Theta(K_{EA}), J(K_{EA}), C(K_{EA}), k(K_{EA})).$$

All symbols above refer exclusively to the corresponding definitions provided by OC core and `ETS.K_EA.v1.1`. No reinterpretation or semantic extension is permitted.

2.1 Representational assumption

To formulate decidability results, we require that an admissible instance of K_{EA} be provided in a *finite*, machine-readable representation consistent with `ETS.K_EA.v1.1`. This assumption does not introduce a new primitive; it merely makes explicit the input model required for any effective decision procedure.

In particular, finiteness applies to the representations of axes, thresholds, flows, cycles, and all auxiliary data structures prescribed by ETS, yielding a finite encoding domain for admissible enterprise states.

2.2 Permitted information for decision procedures

Any decision procedure considered in this work may inspect only the following ETS-provided structures of the given K_{EA} instance:

- the formal representation of admissible states in $\Omega(K_{EA})$ and the associated boundary structure $\partial\Omega(K_{EA})$;
- the finite sets of axes $A(K_{EA})$ and thresholds $\Theta(K_{EA})$;
- any explicitly represented flow and cycle data $(J(K_{EA}), C(K_{EA}))$ contained in the ETS-conformant input;
- the STOP/No-Go constraints and invariants encoded in `ETS.K_EA.v1.1`.

No appeal to managerial intent, KPI observability, or external interpretive semantics is permitted.

2.3 Structure of the formal argument

Within this setting, we proceed as follows: first, structural diagnosability is defined as a Boolean property determined solely by OC and ETS structures; second, this property is shown to be well-defined (invariant under ETS-equivalent representations); and third, decidability is established by reduction to a finite decision problem over the ETS-conformant representation.

3 Definitions

Definition 1 (Admissible enterprise-continuum instance). An *instance* of K_{EA} is an input object that conforms to ETS.K_EA.v1.1 . Such an instance determines, without reinterpretation or semantic enrichment, the corresponding OC core structures $\Omega(K_{\text{EA}})$, $\partial\Omega(K_{\text{EA}})$, $A(K_{\text{EA}})$, $\Theta(K_{\text{EA}})$, $J(K_{\text{EA}})$, $C(K_{\text{EA}})$, and $k(K_{\text{EA}})$, exactly as specified by OC core and ETS.K_EA.v1.1 .

Definition 2 (ETS-induced diagnostic predicates). Fix an admissible instance of K_{EA} . Let $\text{Enc}(K_{\text{EA}})$ denote the finite set of ETS-admissible encodings of enterprise states for this instance, as determined by the finite representations of axes, potentials, thresholds, flows, and cycles prescribed by ETS.K_EA.v1.1 .

The *diagnostic predicate family* $\text{Pred}_D(K_{\text{EA}})$ is the finite set of Boolean predicates over $\text{Enc}(K_{\text{EA}})$ defined as follows:

1. $\text{In}_\Omega(x)$: evaluates to true iff the state decoded from x is an element of $\Omega(K_{\text{EA}})$.
2. $\text{On}_{\partial\Omega}(x)$: evaluates to true iff the state decoded from x lies on the boundary $\partial\Omega(K_{\text{EA}})$.
3. $\text{Sat}_\Theta(x)$: evaluates to true iff the decoded state satisfies all threshold constraints specified by $\Theta(K_{\text{EA}})$ under OC+ETS semantics.
4. $\text{Val}_k(x)$: evaluates to true iff the continuity measure $k(K_{\text{EA}})$ is well-defined at the decoded state and admits evaluation in the numeric domain prescribed by ETS.K_EA.v1.1 .

All predicates in $\text{Pred}_D(K_{\text{EA}})$ are defined solely in terms of OC core structures and ETS.K_EA.v1.1 . No predicate may reference intent, KPI signals, interpretation layers, or external annotations.

Definition 3 (Structural diagnosability). Let K_{EA} be an admissible enterprise-continuum instance. The enterprise continuum K_{EA} is said to be *structurally diagnosable* iff there exists a total decision procedure $\mathcal{A}_D$ such that, for every encoding $x \in \text{Enc}(K_{\text{EA}})$ and every predicate $P \in \text{Pred}_D(K_{\text{EA}})$, the computation $\mathcal{A}_D(P, x)$ halts and returns the correct truth value of $P(x)$ as determined by the OC+ETS.K_EA.v1.1 semantics of the instance.

Structural diagnosability is encoded as the Boolean property

$$D(K_{\text{EA}}) = \begin{cases} 1, & \text{if such a total decision procedure } \mathcal{A}_D \text{ exists,} \\ 0, & \text{otherwise.} \end{cases}$$

Remark 1 (Independence from intent and KPI observability). Independence from managerial intent and KPI observability is enforced by construction. The definition of $D(K_{\text{EA}})$ depends exclusively on the existence of a decision procedure over $\text{Pred}_D(K_{\text{EA}})$, whose inputs and semantics are fully fixed by OC core and ETS.K_EA.v1.1 . No intent variables, KPI signals, or interpretive layers are admissible as inputs.

4 Main Results

Theorem 1 (Well-definedness of structural diagnosability). *Let K_{EA} be an admissible enterprise-continuum instance. The Boolean property $D(K_{EA})$ is well-defined in the following sense: if two concrete representations R_1 and R_2 are equivalent under $ETS.K_EA.v1.1$, then they induce the same value of $D(K_{EA})$.*

Proof sketch. By definition, ETS-equivalent representations induce identical OC core structures $\Omega(K_{EA})$, $\partial\Omega(K_{EA})$, $A(K_{EA})$, $\Theta(K_{EA})$, $J(K_{EA})$, $C(K_{EA})$, and $k(K_{EA})$. Consequently, they induce the same finite encoding domain $\text{Enc}(K_{EA})$ and the same diagnostic predicate family $\text{Pred}_D(K_{EA})$.

The definition of $D(K_{EA})$ depends solely on the existence of a total decision procedure that correctly decides all predicates in $\text{Pred}_D(K_{EA})$ over $\text{Enc}(K_{EA})$. Since both the domain and semantics of these predicates are identical for R_1 and R_2 , the existence (or non-existence) of such a procedure is invariant under ETS-equivalent representation changes. Hence, $D(K_{EA})$ takes the same value for R_1 and R_2 . $\square$

Theorem 2 (Decidability of structural diagnosability). *Let an admissible K_{EA} instance be given by a finite $ETS.K_EA.v1.1$ -conformant representation. Then the structural diagnosability property $D(K_{EA})$ is decidable.*

Proof sketch. By the finiteness requirements implicit in $ETS.K_EA.v1.1$, the encoding set $\text{Enc}(K_{EA})$ is finite, and the diagnostic predicate family $\text{Pred}_D(K_{EA})$ is finite.

For each predicate $P \in \text{Pred}_D(K_{EA})$, its semantics is given by explicit OC/ETS-defined rules over finite data structures (threshold comparisons, boundary membership checks, and evaluation of the continuity measure in the ETS numeric domain). Therefore, the truth value of $P(x)$ is computable for any $x \in \text{Enc}(K_{EA})$.

Structural diagnosability requires the existence of a total decision procedure that decides all such predicate evaluations. Since the domain and predicate set are finite, this existence question reduces to a finite decision problem: one can effectively verify that all predicate evaluations halt and are computable under the given ETS semantics. Hence, $D(K_{EA})$ is decidable. $\square$

5 Scope Limits and Non-Claims

5.1 What is not claimed

We do not claim:

- that diagnosability implies controllability, success, or improvement;
- any managerial prescription or actionability;
- any empirical KPI-based observability guarantee;
- any universality beyond the $ETS.K_EA.v1.1$ and OC scope.

5.2 What would falsify the main claims

The main claims would be falsified if:

- the formal definition of $D(K_{EA})$ cannot be expressed purely in the immutable OC + $ETS.K_EA.v1.1$ structures; or

- the ETS representation necessarily requires infinite objects not reducible to finite decision procedures in the intended encoding.